Radar range and velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate a radar's range measurement from its line-of-sight velocity measurement.

SUCCESS CRITERIA - I can...

1. Use round-trip time for range.
2. Explain the information in a Doppler shift.
3. State what radial velocity leaves unknown.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

An EM wave in vacuum has frequency 5×10^{9} Hz. Use $c=3.00\times10^{8}$ m/s. Find wavelength and name the spectral region.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: range = $c\Delta t/2$; $c = 3.0\times10^{8}$ m/s.
Radar time is a round trip. Doppler gives line-of-sight velocity information.

Worked reasoning: A radar echo returns after 2 microseconds. With $c=3\times10^{8}$ m/s, range$=c\Delta t/2=300$ m. The measured time includes the outward and return paths.

Your turn: Draw the round trip and label the conversion 2 microseconds$=2\times10^{-6}$ s.

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw outgoing and returning radar paths. Which measurement helps range: echo delay or frequency shift? Which helps radial velocity?



Marzano check - circle: ① ② ③

04 Solve and justify

A radar echo returns after .040 s. Find range. A monostatic radar has$\lambda=.10$ m and shift magnitude 400 Hz; use$|v_r|=|\Delta f|\lambda/2$ to find radial speed.

Marzano check - circle: ① ② ③

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05 Analyze the reasoning

An asteroid has zero measured radial velocity at one instant. Does that prove it is stationary? Sketch another possible motion.

Marzano check - circle: 1 2 3

06 Choose a method

A wave travels 12 m in 0.4 s at frequency 60 Hz. Find speed and wavelength.

Marzano check - circle: 1 2 3

07 Exit check

Two targets at the same range have different Doppler shifts. Explain why they can have same time-delay but different motions.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Radar range changes from 1500 to 1440 m over 0.5 s. Find average radial velocity taking outward positive. Explain why total speed may be larger.

Marzano check - circle: 1 2 3